Protein

Ali Sharifi-Zarchi Somayyeh Kouhi

Sharif University of Technology

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Overview

How our body can do such wonderful things?

How our body can do such wonderful things?

When we look at a cell in a microscope or analyze its electrical or biochemical activity, we are, in essence, observing the handiwork of proteins. Proteins are the main building blocks from which cells are assembled, and they constitute most of the cell's dry mass. In addition to providing the cell with shape and structure, proteins also execute nearly all its myriad functions.

The Shape of some Proteins

Figure: Inositol monophosphatase

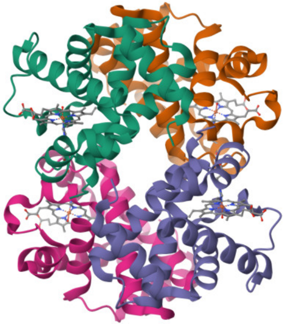


Figure: Greyhound Hemoglobin

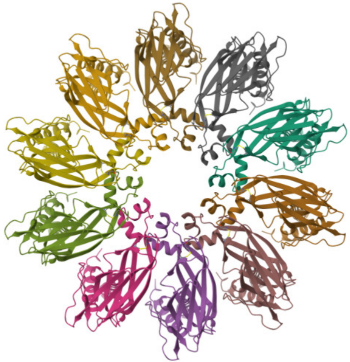


Figure: Seipin

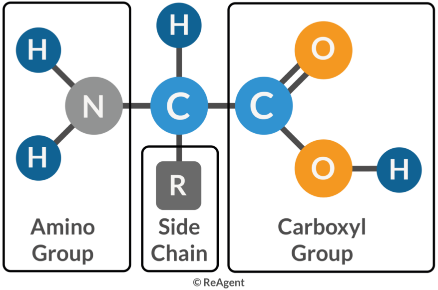
Protein Functions

Proteins exhibit enormous structural complexity and functional sophistication. Major categories include:

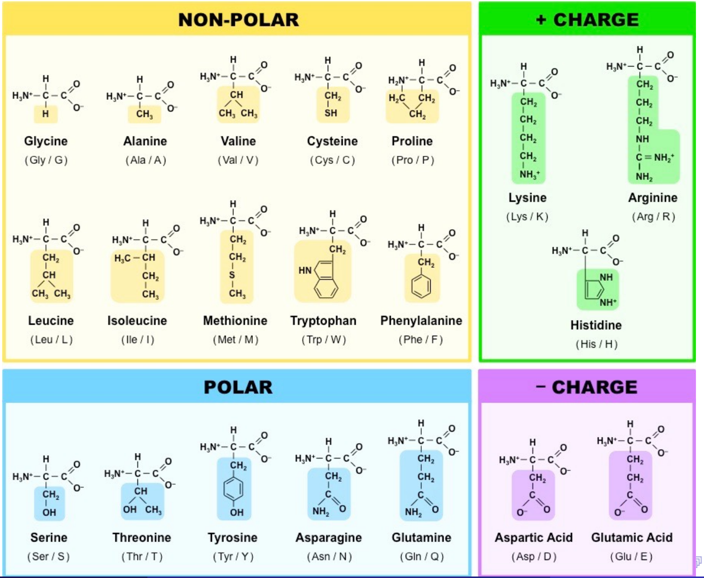
- **Enzymes:** Catalyze biochemical reactions, forming metabolism.
- **Structural Proteins:** Provide mechanical support, e.g., collagen, keratin.
- **Transport Proteins:** Carry molecules or ions, e.g., hemoglobin.
- **Motor Proteins:** Generate movement, e.g., myosin, kinesin.
- **Signal/Receptor Proteins:** Hormones or receptors for communication.
- **Antibodies:** Protect against foreign molecules.
- **Gene Regulatory Proteins:** Switch genes on/off.

Proteins are assembled mainly from a set of 20 different amino acids, each with different chemical properties.

All 20 of the common amino acids are α-amino acids. They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon). They differ from each other in their side chains, or R groups, which vary in structure, size, and electric charge, and which influence the solubility of the amino acids in water.



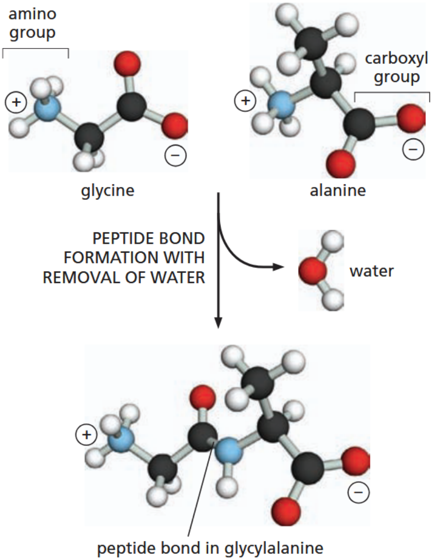
20 Amino Acids



Amino acids are linked together by peptide bonds

A protein molecule is made from a long chain of these amino acids, held together by covalent peptide bonds. Proteins are therefore referred to as polypeptides, or polypeptide chains. In each type of protein, the amino acids are present in a unique order, called the amino acid sequence, which is exactly the same from one molecule of that protein to the next.

Amino acids are linked together by peptide bonds



Question

We understand what a protein is

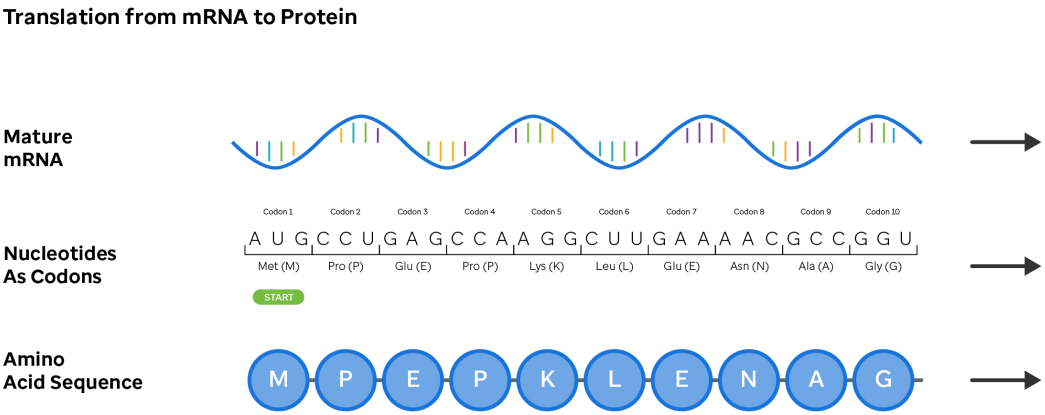
the next question is: how are proteins built?

An mRNA Sequence Is Decoded in Sets of Three Nucleotides

Transcription, the transfer of information from DNA to RNA, is simple to understand as DNA and RNA are chemically and structurally similar, allowing DNA to act as a direct template for RNA synthesis through complementary base-pairing. This process is analogous to converting a handwritten message into a typewritten one, where the language and symbols remain closely related. In contrast, the conversion of

information from RNA into protein represents a **Translation into another language** using different symbols. This translation cannot be a direct one-to-one correspondence, as there are only 4 different nucleotides in mRNA but 20 different types of amino acids in a protein. The set of rules by which the nucleotide sequence of a gene is translated into the amino acid sequence of a protein, via an intermediary mRNA molecule, is known as the genetic code.

An mRNA Sequence Is Decoded in Sets of Three Nucleotides



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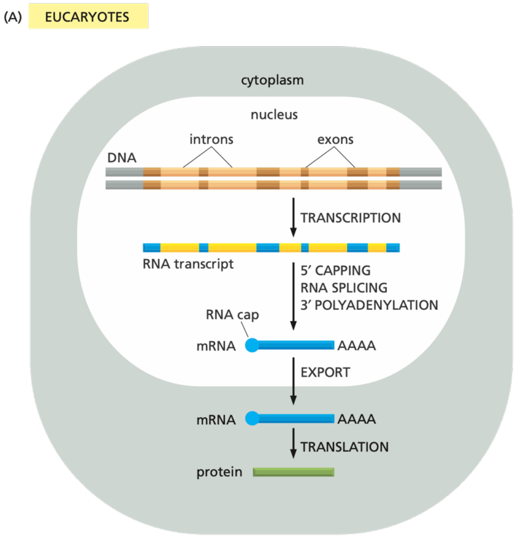
Codon

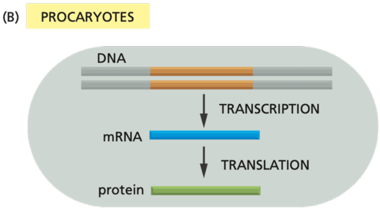
In 1961, it was discovered that the sequence of nucleotides in an mRNA molecule is read consecutively in groups of three. And because RNA is made of 4 different nucleotides, there are $4 \times 4 \times 4 = 64$ possible combinations of three nucleotides: AAA, AUA, AUG, and so on. However, only 20 different amino acids are commonly found in proteins. Either some nucleotide triplets are never used, or the code is redundant, with some amino acids being specified by more than one triplet. The second possibility turned out to be correct, as shown by the completely deciphered genetic code shown in next slide picture. Each group of three consecutive nucleotides in RNA is called a **codon**, and each codon specifies one amino acid.

Codons

codons	GCA GCC GCG GCU	AGA AGG CGA CGC CGG CGU	GAC GAU	AAC AAU	UGC UGU	GAA GAG	CAA CAG	GGA GGC GGG GGU	CAC CAU	AUA AUC AUU	UUA UUG CUA CUC CUG CUU	AAA AAG	AUG	UUC UUU	CCA CCC CCG CCU	AGC AGU UCA UCC UCG UCU	ACA ACC ACG ACU	UGG	UAC UAU	GUA GUC GUG GUU	UAA UAG UGA
amino acids	Ala	Arg	Asp	Asn	Cys	Glu	Gln	Gly	His	Ile	Leu	Lys	Met	Phe	Pro	Ser	Thr	Trp	Tyr	Val	stop
	A	R	D	N	C	E	Q	G	H	I	L	K	M	F	P	S	T	W	Y	V	

Eukaryotic VS Prokaryotic





Eukaryotes vs Prokaryotes

(A) Eukaryotes

In eukaryotic cells, the pre-mRNA molecule from transcription contains introns and exons. Its ends undergo capping and polyadenylation, and introns are removed by RNA splicing. The finished mRNA is transported from the nucleus to the cytosol for translation.

These steps are depicted sequentially but often occur simultaneously; for instance, capping and splicing typically start before transcription finishes. Consequently, full transcripts of the entire gene (including all introns) usually do not exist in the cell.

Finally, mRNAs are degraded in the cytosol by RNases, and their nucleotides are reused.

(B) Prokaryotes

In prokaryotes, mRNA production is simpler. Transcription initiation by RNA polymerase produces the 5 end, and transcription termination produces the 3 end.

Because there is no nucleus, transcription, translation, and degradation all occur in a common compartment. Therefore, translation of a prokaryotic mRNA can start before its synthesis is finished.

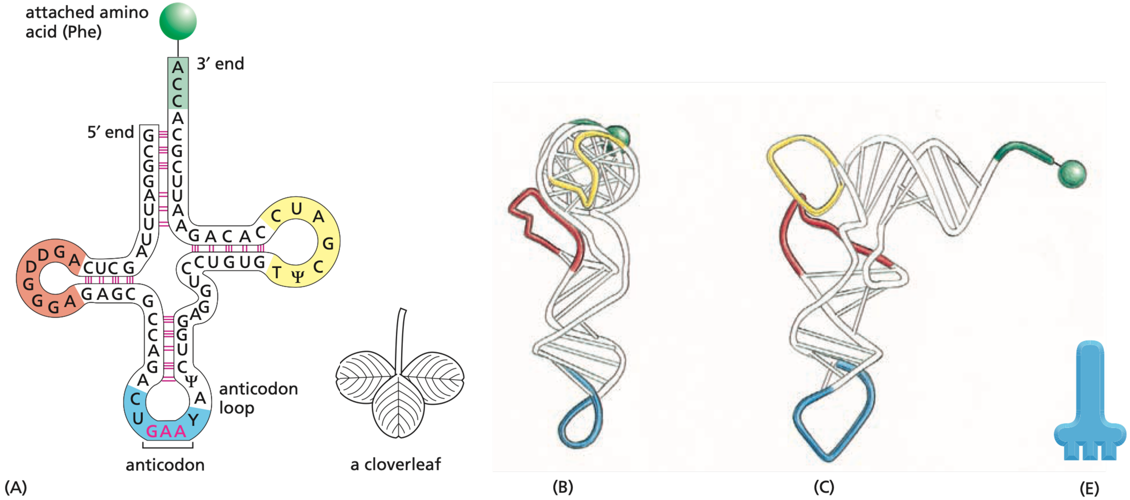
In both cell types, the amount of a protein is determined by the rates of each of these steps, as well as the degradation rates of the mRNA and protein molecules.

Translation: Beyond the Basics

We've understood the main idea of translation.

Now let's take a deeper look at its mechanism.

tRNA structure



Specific Enzymes Couple tRNAs to the Correct Amino Acid

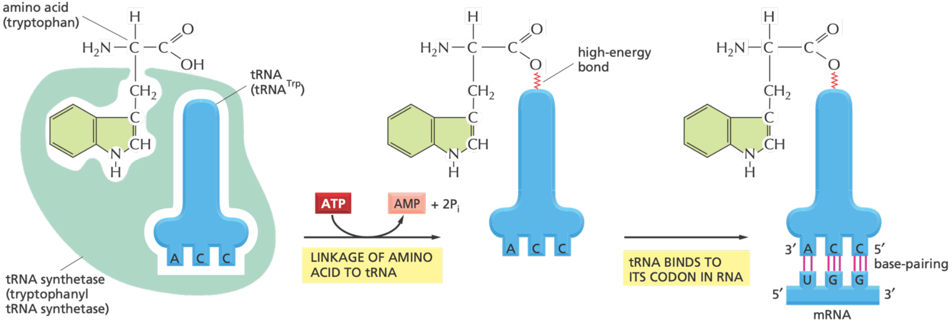
For a tRNA molecule to carry out its role as an adaptor, it must be linked or charged with the correct amino acid. How does each tRNA molecule recognize the one amino acid in 20 that is its proper partner? Recognition and attachment of the correct amino acid depend on enzymes called aminoacyl-tRNA synthetases, which covalently couple each amino acid to the appropriate set of tRNA molecules.

Specific Enzymes Couple tRNAs to the Correct Amino Acid

In most organisms, there is a different synthetase enzyme for each amino acid.Each synthetase enzyme recognizes its designated amino acid, as well as nucleotides in the anticodon loop and in the amino-acid-accepting arm that are specific to the correct tRNA.

The synthetase-catalyzed reaction that attaches the amino acid to the 3 end of the tRNA. The reaction produces a high-energy bond between the charged tRNA and the amino acid. The energy of this bond is later used to link the amino acid covalently to the growing polypeptide chain.

The genetic code is translated by aminoacyl-tRNA synthetases and tRNAs



Ribosomes

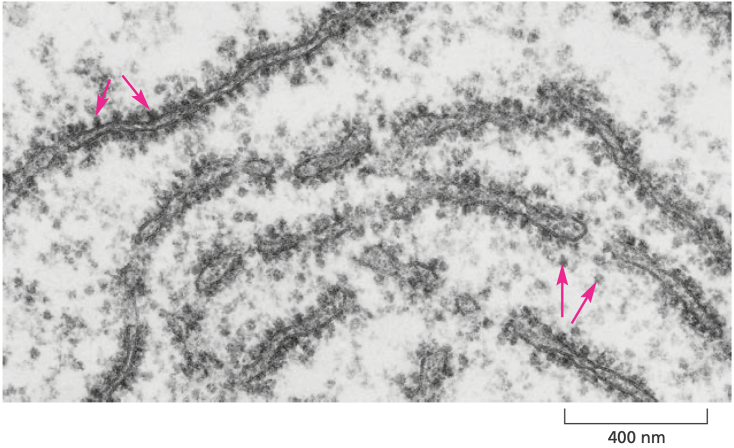
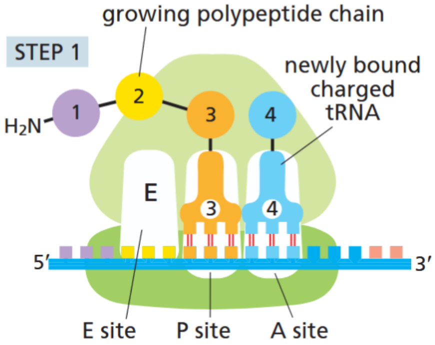


Figure: The mRNA Message Is Decoded on Ribosomes

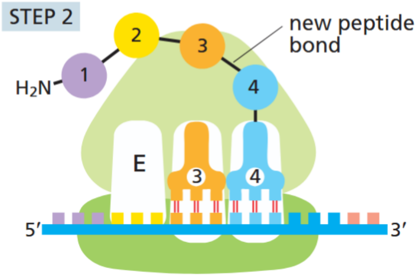
Translation takes place in a four-step cycle

In step 1, a charged tRNA carrying the next amino acid to be added to the polypeptide chain binds to the vacant A site on the ribosome by forming base pairs with the mRNA codon that is exposed there. Only a matching tRNA molecule can base-pair with this codon, which determines the specific amino acid added. The A and P sites are sufficiently close together that their two tRNA molecules are forced to form base pairs with codons that are contiguous, with no stray bases in-between. This positioning of the tRNAs ensures that the correct reading frame will be preserved throughout the synthesis of the protein.



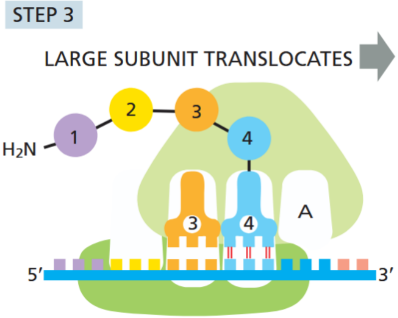
Step 2

In step 2, the carboxyl end of the polypeptide chain (amino acid 3 in step 1) is uncoupled from the tRNA at the P site and joined by a peptide bond to the free amino group of the amino acid linked to the tRNA at the A site. This reaction is carried out by a catalytic site in the large subunit.

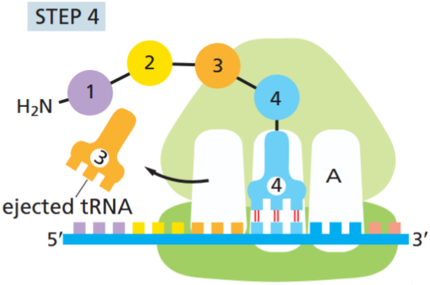


Step 3

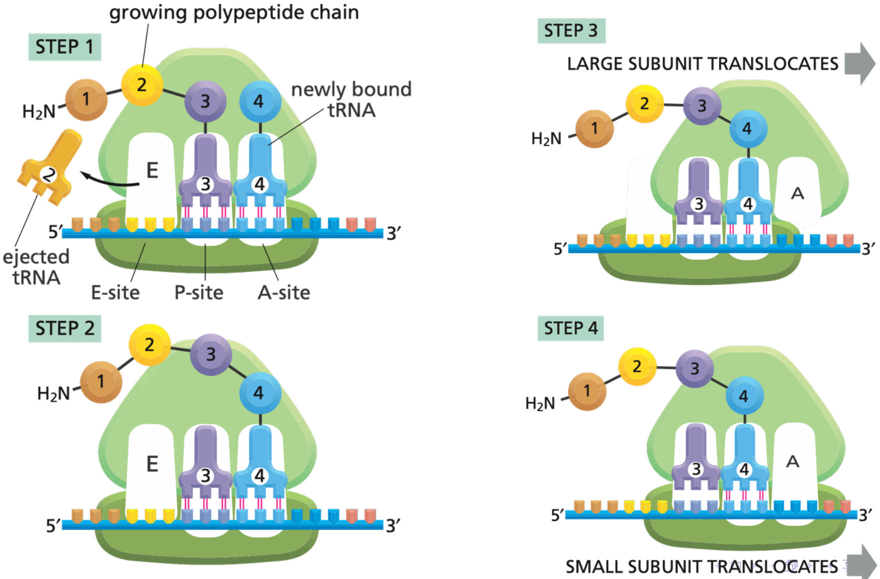
In step 3, a shift of the large subunit relative to the small subunit moves the two bound tRNAs into the E and P sites of the large subunit.



In step 4, the small subunit moves exactly three nucleotides along the mRNA molecule, bringing it back to its original position relative to the large subunit. This movement ejects the spent tRNA and resets the ribosome with an empty A site so that the next charged tRNA molecule can bind.

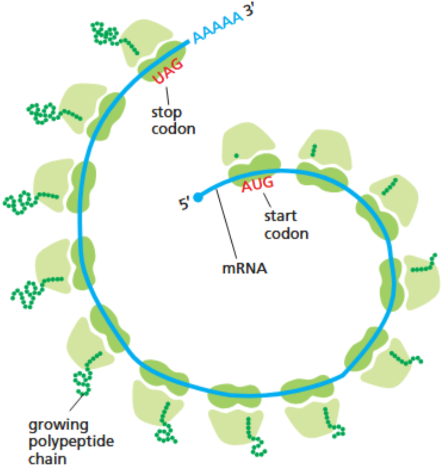


4-Steps


STEP 1
growing polypeptide chain
newly bound
tRNA
ejected
tRNA
E-site
P-site
A-site
STEP 2
STEP 3
LARGE SUBUNIT TRANSLOCATES
STEP 4
SMALL SUBUNIT TRANSLOCATES

Proteins Are Produced on Polyribosomes

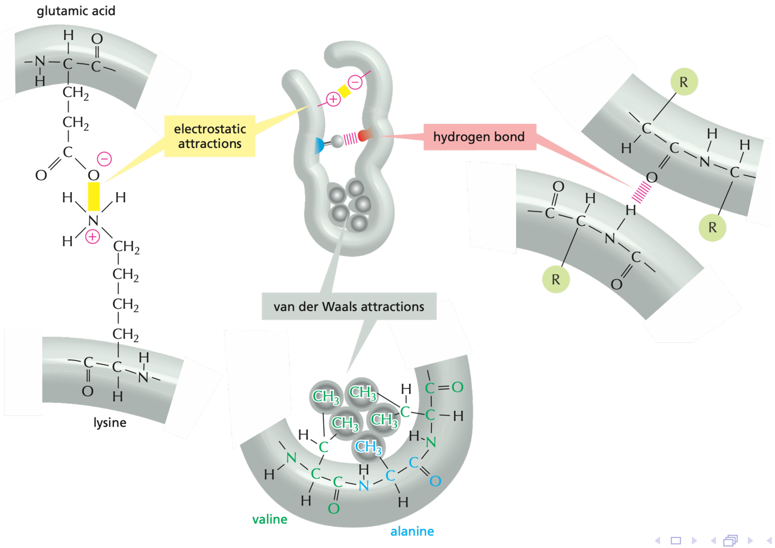
The synthesis of most protein molecules takes between 20 seconds and several minutes. But even during this short period, multiple ribosomes usually bind to each mRNA molecule being translated. If an mRNA is being translated efficiently, a new ribosome will hop onto its 5 end almost as soon as the preceding ribosome has translated enough of the nucleotide sequence to move out of the way. The mRNA molecules being translated are therefore usually found in the form of polyribosomes, also known as **polysomes**. These large cytosolic assemblies are made up of many ribosomes spaced as close as 80 nucleotides apart along a single mRNA molecule



Protein Structures

we know the Amino Acid sequences, is it enough?

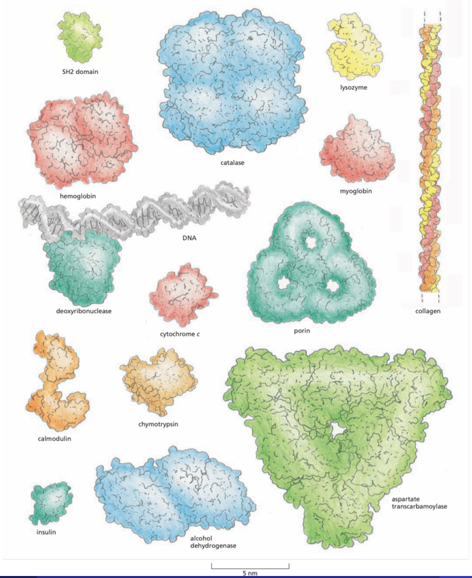
Non-covalent Bons



Non-covalent Bons

Because a noncovalent bond is much weaker than a covalent bond, it takes many noncovalent bonds to hold two regions of a polypeptide chain tightly together. The stability of each folded shape is largely determined by the combined strength of large numbers of noncovalent bonds

Proteins Come in a Wide Variety of Complicated Shapes



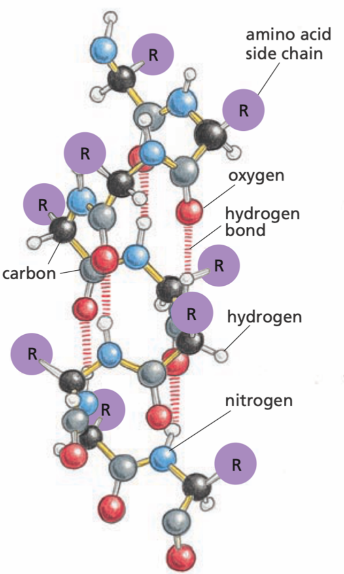
Proteins Come in a Wide Variety of Complicated Shapes

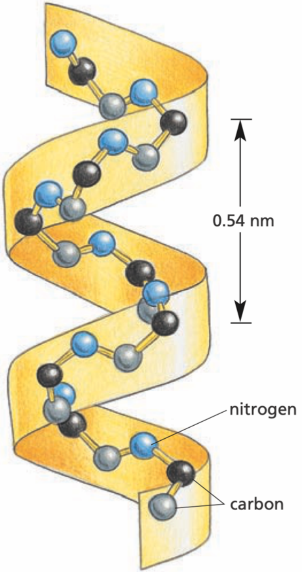
Each protein normally folds into a single, stable conformation. This conformation, however, often changes slightly when the protein interacts with other molecules in the cell. Such changes in shape are crucial to the function of the protein, as we discuss later.

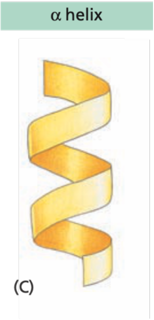
The α Helix and the β Sheet Are Common Folding Patterns

Because the amino acid side chains are not involved in forming these hydrogen bonds, helices and sheets can be generated by many different amino acid sequences.

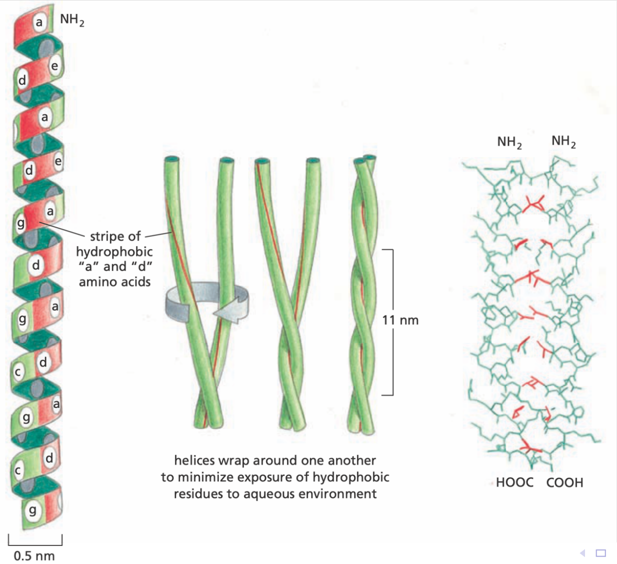
α-helix


amino acid side chain
R
oxygen
hydrogen bond
carbon
hydrogen
nitrogen

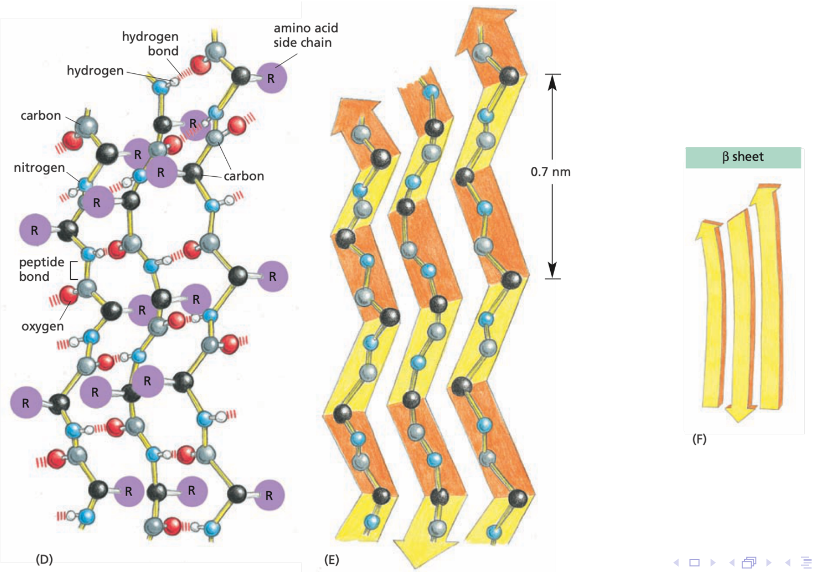

0.54 nm
nitrogen
carbon


α helix
(C)

Coiled-coil Helices

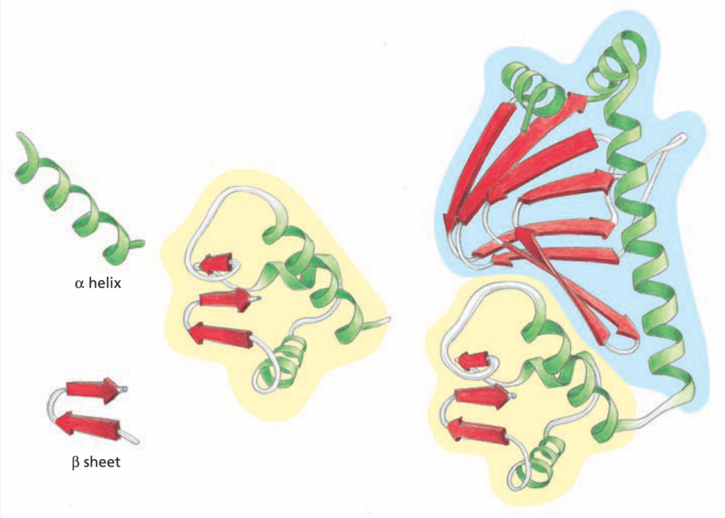


β-Sheets



hydrogen bond
amino acid side chain
hydrogen
carbon
nitrogen
carbon
peptide bond
oxygen
R
(D)
(E)
0.7 nm
β sheet
(F)

Different Structures

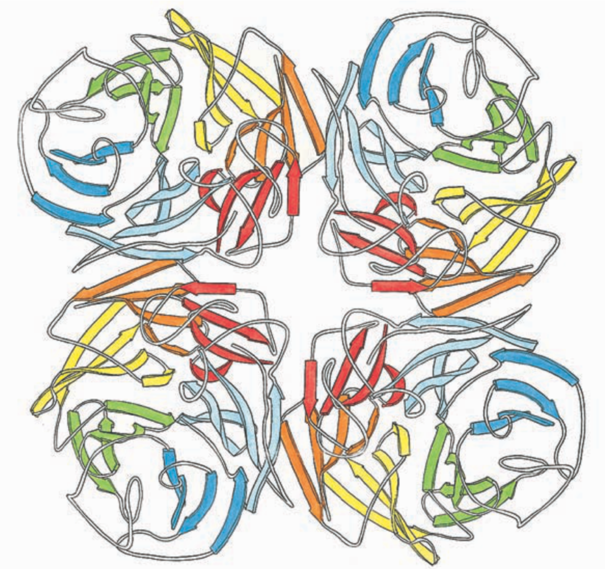


secondary structure

single polypeptide domain

protein molecule made of two different domains

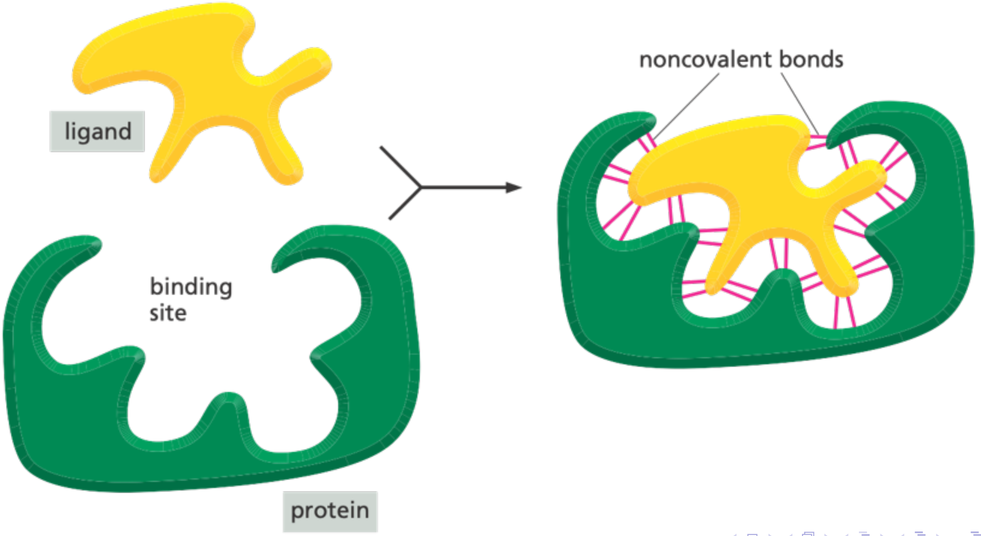
Repeated Units



Protein Binding

Proteins can build larger structures by repeatedly linking identical subunits, forming filaments, sheets, or spherical assemblies. When one protein has a binding site that matches a specific region on another protein of the same kind, they can join in a repeating pattern. Because each interaction occurs in the same way, the resulting assembly often takes on a helical structure that can extend indefinitely in both directions.

Protein Binding



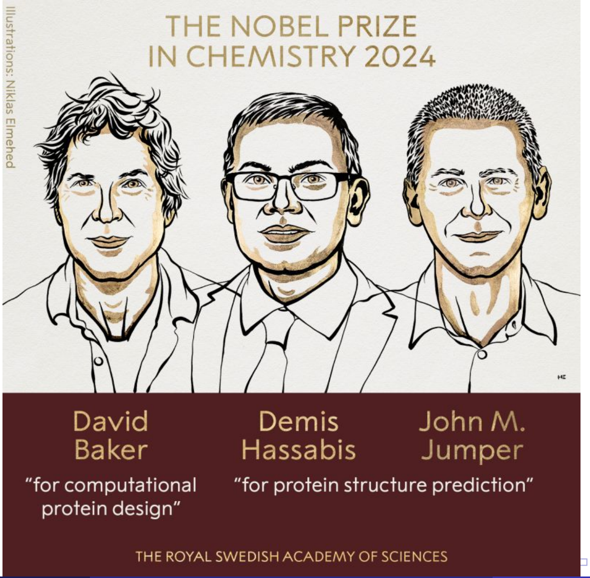
Protein Folding and AlphaFold

The challenge was a 50-year biological puzzle: determining a protein's 3D structure solely from its amino-acid sequence.

AlphaFold

In 2021, DeepMind introduced AlphaFold, an AI system that solved a 50-year-old biological challenge: predicting the 3D structure of proteins from their amino acid sequences. By leveraging deep learning and neural networks, AlphaFold achieved unprecedented accuracy in protein structure prediction. Its architecture includes two main modules: an evolutionary multiple sequence alignment (MSA) module, which extracts evolutionary relationships among sequences, and a structure module, which predicts the protein's 3D coordinates. This breakthrough was celebrated globally, earning its developers recognition akin to a Nobel-level achievement in biology.

AlphaFold


Illustrations: Niklas Elmehed
THE NOBEL PRIZE
IN CHEMISTRY 2024
David
Baker
"for computational
protein design"
Demis
Hassabis
John M.
Jumper
"for protein structure prediction"
THE ROYAL SWEDISH ACADEMY OF SCIENCES

Protein Databases

What Are Protein Databases?

What Are Protein Databases?

Protein databases store and organize experimental and computational data about proteins. They are indispensable for:

- Sequence retrieval and comparison.
- Functional annotation.
- Structural modeling.
- Drug discovery and target identification.

Primary vs Secondary Databases

- **Primary Databases:** Contain experimentally determined data.
  - *UniProtKB, PDB, GenPept.*
- **Secondary Databases:** Derive information from primary data (annotations, motifs, families).
  - *Pfam, InterPro, PROSITE, SMART.*

Major Protein Databases

- **UniProtKB:** Comprehensive protein sequence and functional information.
- **PDB (Protein Data Bank):** 3D structures from X-ray, NMR, and cryo-EM.
- **InterPro:** Integrates multiple signature databases for domain analysis.
- **Pfam:** Protein families and hidden Markov models (HMMs).
- **PROSITE:** Database of motifs and patterns.
- **AlphaFold DB:** AI-predicted structures of millions of proteins.

Example: UniProt Entry

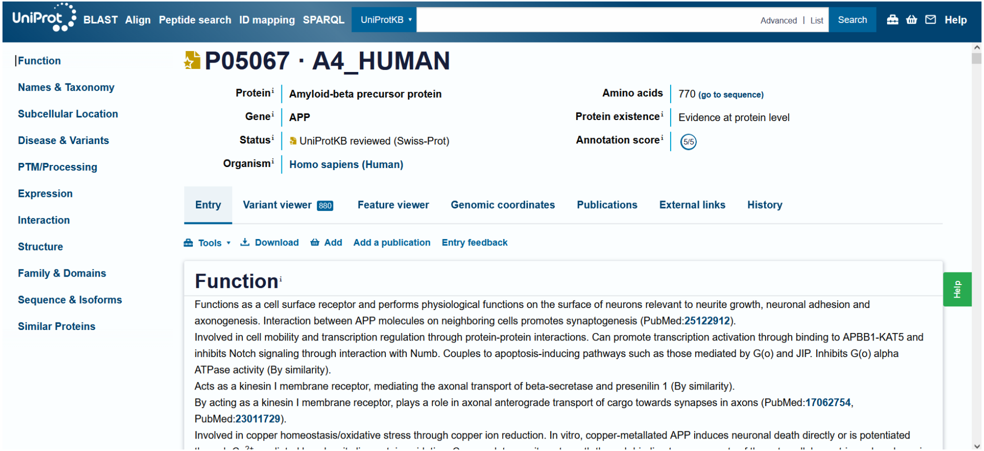


Figure: Example of a UniProtKB protein entry showing sequence, function, and domain information.

Integrating Data from Multiple Sources

Protein databases are often cross-linked:

- UniProt $\leftrightarrow$ PDB (structure links).
- UniProt $\leftrightarrow$ Pfam/InterPro (domains).
- UniProt $\leftrightarrow$ GO (Gene Ontology annotations).

Goal: Enable unified access to multi-dimensional data — sequence, structure, and function.

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The End

Questions? Comments?